

SURYAPRAKASH B

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CAREER OBJECTIVE

Industrious final year electrical engineering student with the high motive of getting into a challenging environment, where I exhibit my skills and strengths by being presented as a diligent resource, thereby enhancing my knowledge and future career.

EDUCATION

B.Tech in Electrical and Electronics Engineering (2022 – 2026)

Sri Manakula Vinayagar Engineering College – 6.87(up to 6th semester)

Higher Secondary School Examination (2022)

Model school – 66%

Secondary School Examination (2020)

Sri Vidya Mandir Senior Secondary School – 64%

SOFT SKILLS

- Time management
- Good in Teamwork
- Adaptability
- Leadership
- Active listening

TECHNICAL SKILLS

- Verilog HDL
- C
- Proteus
- MATLAB
- FPGA

ACHIEVEMENTS

- Participated in Paper Presentation about Smart Vehicle Monitoring System (SVMS) held at Coimbatore Institute of Technology (CIT, Tamil Nadu).
- Participated in IDE Bootcamp Edition-II Phase-1 conducted by AICTE & MIC at Amal Jyothi College of Engineering (Kottayam, Kerala).

MINI PROJECTS

- **RC car using Bluetooth module:** Developed a Bluetooth-controlled RC car using a microcontroller and Bluetooth module, demonstrating proficiency in embedded systems and wireless communication. Successfully implemented circuit and code for remote control via smartphone app.
- **Anti sleep alarm detector:** Developed an Anti-Sleep Alarm Detector using sensors and

microcontrollers to detect driver drowsiness and alert them with alarms, enhancing road safety and reducing accidents caused by driver fatigue.

- **Water level indicator:** Designed and developed a Water Level Indicator using sensors and microcontrollers to monitor water levels in tanks, providing real-time alerts when levels reach predefined thresholds, and helping conserve water and prevent overflow.

CERTIFICATIONS

- Introduction in PCB (NPTEL)
- Introduction To Internet of Things (NPTEL)
- Electric Vehicle Battery Management (Udemy)
- MATLAB On RAMP

AREA OF INTEREST

- Power Electronics: Rectifiers and AC voltage controller
- Programming in C
- Programming in PYTHON
- Cloud Computing
- PCB Designing

HOBBIES

- Cricket
- Chess
- Foot Ball

LANGUAGES

- English
- Tamil
- Hindi

WORKSHOPS AND WEBINARS

- **Blockchain Workshop:** MVJ College of Engineering, Bangalore
- **IDE Bootcamp:** Amal Jyothi College of Engineering, Kottayam, Kerala

FINAL YEAR PROJECT

- **ONBOARD CHARGER BY USING MPCC IN EV:** In the evolving landscape of electric mobility, efficient and intelligent charging solutions play a pivotal role in enhancing the reliability and sustainability of plug-in electric vehicles (EVs). This project presents a Onboard Charger (OBC) system featuring Model Predictive Current Control (MPCC) to optimize the performance of EV charging operations. The proposed architecture incorporates an AC supply interface, EMI filter, and an interleaved boost Power Factor Correction (PFC) converter, followed by a DC link capacitor for energy stabilization. A bidirectional full-bridge DC-DC converter is employed to facilitate power flow in both charging and regenerative modes, making the system suitable for vehicle-to-grid (V2G) applications.
- The MPCC algorithm leverages real-time feedback from voltage and current sensors to ensure high dynamic performance, minimized total harmonic distortion (THD), and enhanced control precision...